COURSE DESCRIPTION FORM: CS-3009: Intro to Software Engineering

INSTITUTION FAST School of Computing, National University of Computer and Emerging Sciences, Islamabad Campus

BS-CS: **Spring-2026**

PROGRAM(s) TO BE EVALUATED

Course Description

<table>
<tr><td>Course Code</td><td>CS-3009</td></tr>
<tr><td>Course Title</td><td>Software Engineering</td></tr>
<tr><td>Credit Hours</td><td>3</td></tr>
<tr><td>Course Instructors</td><td>Dr. Labiba Fahad</td></tr>
<tr><td>Grading Policy</td><td>Absolute Grading</td></tr>
<tr><td>Policy about missed assessment items in the course</td><td>Retake of missed assessment items (other than sessional/ final exam) will not be held. Student who misses an assessment item (other than sessional / final exam) is awarded zero marks in that assessment item i.e., late submission will not be accepted.
For missed sessional/ final exam, exam retake/ pretake application along with necessary evidence are required to be submitted to the department secretary. The examination assessment and retake committee decides the exam retake/ pretake cases.</td></tr>
<tr><td>Course Plagiarism Policy</td><td>Plagiarism in project or sessional/ final exam will result in F grade in the course.
Plagiarism in an assignment will result in zero marks in the whole assignments category.</td></tr>
<tr><td>Prerequisites by Course(s) or Topics</td><td></td></tr>
<tr><td>Assessment Instruments with Weights (homework, quizzes, sessional exams, final exam, assignments, etc.)</td><td>Assessment with weight.
<table>
<tr><th>Assessment Type</th><th>Weight</th></tr>
<tr><td>Quiz</td><td>7.5</td></tr>
<tr><td>Assignment</td><td>7.5</td></tr>
<tr><td>Class participation</td><td>5</td></tr>
<tr><td>Sessional Exams 1</td><td>15</td></tr>
<tr><td>Sessional Exams 2</td><td>15</td></tr>
<tr><td>Project</td><td>10</td></tr>
<tr><td>Final Exam</td><td>40</td></tr>
</table></td></tr>
<tr><td>Course Coordinator</td><td>Dr. Labiba Fahad</td></tr>
<tr><td>URL (if any)</td><td></td></tr>
<tr><td>Course Catalog Description</td><td>This course introduces students to the fundamental principles and methodologies of Software Engineering. Course will cover traditional and modern software development life</td></tr>
</table>

	cycles' stages with practical examples and case studies. Furthermore, we will cover RE, Design, architecture, testing, and project management in detail. I have introduced new topics in all of the above mentioned traditional workflow of the software development process. I will also cover software process improvement models and their details in the course.
Textbook(s)	**Software Engineering, Ninth Edition, 2010. Sommerville, Ian Addison Wesley**
Reference Material	SE and Testing, b. B. Agarwal s. P. Tayal m. Gupta, Jones and Bartlett Publishers. Software Engineering: A Practitioner's Approach, Pressman, R.S. & Maxim B., 8th Edition (2015), McGraw-Hill.
Course Goals	

A. Course Learning Outcomes (CLOs)

After course completion, the students shall be able to:
1. Select an appropriate software development process for a software project (4) (3)
2. Develop a model of requirements for a software system (6) (4)
3. Design architecture of a software system by choosing the most appropriate architecture styles (6) (4)
4. Design test cases for a software system (6) (4)
5. Construct reasonable sized software in team setting (6) (4)

B. Program Learning Outcomes (PLOs)

PLO 1	Academic Education	Completion of an accredited program of study designed to prepare graduates as computing professionals
PLO 2	Knowledge for Solving Computing Problems	Apply knowledge of computing fundamentals, knowledge of a computing specialization, and mathematics, science, and domain knowledge appropriate for the computing specialization to the abstraction and conceptualization of computing models from defined problems and requirements
PLO 3	Problem Analysis	Identify, formulate, research literature, and solve complex computing problems reaching substantiated conclusions using fundamental principles of mathematics, computing sciences, and relevant domain disciplines
PLO 4	Design/ Development of Solutions	Design and evaluate solutions for complex computing problems, and design and evaluate systems, components, or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations

PLO 5	Modern Tool Usage	Create, select, adapt and apply appropriate techniques, resources, and modern computing tools to complex computing activities, with an understanding of the limitationsCreate, select, adapt and apply appropriate techniques, resources, and modern computing tools to complex computing activities, with an understanding of the limitations
PLO 6	Individual and Team Work	Function effectively as an individual and as a member or leader in diverse teams and in multi-disciplinary settings
PLO 7	Communication	Communicate effectively with the computing community and with society at large about complex computing activities by being able to comprehend and write effective reports, design documentation, make effective presentations, and give and understand clear instructions
PLO 8	Computing Professionalism and Society	Understand and assess societal, health, safety, legal, and cultural issues within local and global contexts, and the consequential responsibilities relevant to professional computing practice
PLO 9	Ethics	Understand and commit to professional ethics, responsibilities, and norms of professional computing practice
PLO 10	Life-long Learning	Recognize the need, and have the ability, to engage in independent learning for continual development as a computing professional

C. Mapping of CLOs to PLOs
(CLO: Course Learning Outcome, PLOs: Program Learning Outcomes)

		1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
CLOs	1			✓							
	2				✓						
	3				✓						
	4				✓						

	5			✓							

Topics covered in the course (assume 15-week instruction and 3 contact hours per week)	**Topics to be covered:**

List of Topics	No. of Weeks	Contact Hours	CLO(s)
Introduction to Software Engineering	1	3	1
Software Process Models (traditional models)	2	6	1,2,6
Software Process Models (non-traditional models)	2	6	1,2,6
Requirement Engineering	2	6	2
Software Architecture design	2.5	7.5	1,3,4,6
Software Testing Basics	2	6	1,3,6
Software Quality Assurance and Processes	1.5	4.5	3
Software Project Management (Planning, Costing, Scheduling)	2	6	1,5
Total	**15**	**45**	

Programming Language for Assignments (if any)	C++ and JAVA			
Class Time Spent (in percentage)	**Theory**	**Problem Analysis**	**Solution Design**	**Social and Ethical Issues**
	50	25	20	5
Oral and Written Communications	Every student is required to submit at least __5___ written reports of typically _4_ pages and make __1___ oral presentation of typically ____20___ minutes' duration.			